

Column	Type	Null	Default	Comments
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Date_Issued	date	No		
certificate_path	varchar(255)	Yes	NULL	

Achievement_ID	userID	Subject_Code	Date_Issued	certificate_path
1	D032410025	DITP2913	2026-07-02	

Column	Type	Null	Default	Comments
Chapter_ID	int(11)	No		
Subject_Code	varchar(20)	No		
Chapter_Name	varchar(100)	No		
chapter_order	int(11)	Yes	1	

Chapter_ID	Subject_Code	Chapter_Name	chapter_order
1	DITP2913	Intro to database design	1
2	DITP2913	chapter 2- Query	1
3	DITP1113	INTRODUCTION TO C++	1
4	DITP1113	Control Structure	1
5	DITP1113	Loop in C++	1
6	DITS1133	Basic of Computer Architectur	1
7	DITS1133	PROCESSOR ARCHITECTURE	1
8	DITS1133	MEMORY SYSTEM	1
16	DITP2913	1	1
17	DITP2913	2	1
18	DITP1113	1	1
19	DITP1113	2	1
20	DITP1113	3	1
21	DITS1133	1	1
22	DITS1133	2	1
23	DITS1133	3	1

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Chapter_ID	int(11)	No		
Name	varchar(100)	No		
File_Type	varchar(50)	No		
file_path	varchar(255)	No		

Content_ID	Chapter_ID	Name	File_Type	file_path
4	1	Intro to database design	slide	uploads/1782891457_1782890339_Lecture4_Function_Part1.pdf
7	1	Intro to database video	video	https://www.youtube.com/watch?v=wR0jg0eQsZA
8	2	chapter 2- Query	slide	uploads/1782929868_Lecture5_Function_Part2.pdf
9	2	QUERY	video	https://www.youtube.com/watch?v=LUL1nnxUz_c&t=24s
18	3	INTRODUCTION TO C++	video	https://www.youtube.com/watch?v=s0g4ty29Xgg
19	4	Control Structure	video	https://www.youtube.com/watch?v=jzgTxosgGIA
20	5	Loop in C++	video	https://www.youtube.com/watch?v=jNI5gj_xSNQ
21	6	Basic of Computer Architectur	video	https://www.youtube.com/watch?v=GRInNLx3Tug
22	7	PROCESSOR ARCHITECTURE	video	https://www.youtube.com/watch?v=GtVDTp826DE
23	8	MEMORY SYSTEM	video	https://www.youtube.com/watch?v=PujjqfUhtNo&list=PLBlnK6fEyqRjdT1xkkBZSXXwFKqQoYhwy

Column	Type	Null	Default	Comments
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Subject_Code	varchar(20)	No		

userID	Subject_Code
D032410021	DITP2913
D032410025	DITP1113
D032410025	DITP2913

Column	Type	Null	Default	Comments
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Subject_Code	varchar(20)	No		

userID	Subject_Code
L1006001	DITP2913
L1006789	DITS1133
L1006893	DITP1113

Column	Type	Null	Default	Comments
Quiz_ID	int(10)	No		
Chapter_ID	int(11)	Yes	NULL	
Quiz_title	varchar(100)	No		
Passing_Mark	int(3)	No		

Quiz_ID	Chapter_ID	Quiz_title	Passing_Mark
1	1	CH01 DATABASE	70
2	1	CH02 DATABASE	70
3	1	CH03 DATABASE	70
4	1	CH04 DATABASE	70
5	1	CH05 DATABASE	70
101	3	Chapter 1: Introduction to C++	70
102	3	Chapter 2: Control Structures	70
103	3	Chapter 3: Loops and Iterations	70
104	3	Chapter 4: Functions	70
105	3	Chapter 5: Arrays	70
201	6	Chapter 1: Computer Basics	70
202	6	Chapter 2: Processor Architecture	70
203	6	Chapter 3: Memory Systems	70
204	6	Chapter 4: I/O Systems	70
205	6	Chapter 5: Pipelining	70

Column	Type	Null	Default	Comments
Question_ID	int(11)	No		
Quiz_ID	int(11)	No		
Question_Text	text	No		
OptA	varchar(255)	No		
OptB	varchar(255)	No		
OptC	varchar(255)	No		
OptD	varchar(255)	No		
Correct_Answer	varchar(1)	No		

Question_ID	Quiz_ID	Question_Text	OptA	OptB
4	1	What does SQL stand for?	Structured Query Language	Sequential Query Logic
5	1	Which of the following is an example of a Relational Database Management System (RDBMS)?	MongoDB	MySQL
6	1	In a relational database, what is a single row in a table called?	Attribute	Tuple (or Record)
7	1	What database language is used to retrieve and manipulate data (e.g., SELECT, INSERT, UPDATE)?	DDL (Data Definition Language)	DML (Data Manipulation Language)
8	1	Which SQL clause is used to filter results based on a specific condition?	ORDER BY	GROUP BY
9	1	What is a primary key?	A unique identifier for each record in a table	A key used to encrypt database files
10	1	What is a foreign key?	A key that is used to unlock a database from another server	A field in one table that uniquely identifies a row of another table
11	1	Which of the following is NOT a core property of a relational transaction (ACID)?	Atomicity	Consistency
12	1	What is the process of organizing data in a database to reduce redundancy and improve data integrity called?	Normalization	Optimization
13	1	What type of command is used to change the structure of a database (like adding a new column to a table)?	SELECT	ALTER
16	2	Which SQL statement is used to retrieve data from a database table?	GET	SELECT
17	2	Which symbol is used in the SELECT statement to retrieve all columns from a table?	*	&
18	2	Which clause is used to filter the results and only retrieve rows that meet a specific condition?	HAVING	WHERE
19	2	To sort the result set in ascending or descending order, which clause is used?	SORT BY	ORDER BY
20	2	Which SQL keyword is used to return only unique (different) values, removing duplicates?	DIFFERENT	DISTINCT
21	2	Which keyword or function is used to calculate the total number of rows returned by a query?	TOTAL()	COUNT()
22	2	If you only want to retrieve a limited number of rows (e.g., the top 5), which clause do you use?	LIMIT	MAX
23	2	When combining data from two or more tables based on a related column, which clause is used?	MERGE	JOIN
24	2	Which clause is used to group rows that have the same values into summary rows, often used with aggregate functions?	GROUP BY	SUMMARIZE
25	2	What is the correct syntax to retrieve only the 'FirstName' and 'LastName' columns from a 'Employees' table?	SELECT * FROM Employees WHERE FirstName, LastName	SELECT FirstName, LastName FROM Employees
26	3	Which keyword is used to combine rows from two or more tables based on a related column between them?	COMBINE	JOIN
27	3	What type of JOIN returns only the rows that have matching values in both tables?	INNER JOIN	LEFT JOIN
28	3	In a SQL query, where should the ORDER BY clause typically be placed?	Before the FROM clause	Between the FROM and JOIN clauses
29	3	Which keyword is used with ORDER BY to sort the retrieved data from highest to lowest value?	ASC	DESC
30	3	If you do not specify ASC or DESC in an ORDER BY clause, how will the database sort the results by default?	In descending order	In ascending order
31	3	Look at the following query syntax. What is missing to correctly correlate the tables? SELECT orders.order_id, customers.name FROM orders INNER JOIN customers _____ orders.customer_id = customers.customer_id;	ON	WHERE
32	3	Which of the following statements correctly combines two tables and sorts the output by employee names alphabetically?	SELECT * FROM employees JOIN departments ON employees.dept_id = departments.dept_id ORDER BY employee_name;	SELECT * FROM employees JOIN departments WHERE employees.dept_id = departments.dept_id ORDER BY employee_name DESC;
33	3	If you want to join an 'orders' table (left) with a 'customers' table (right) and return all records from the orders table even if there is no match in the customers table, which join should you use?	RIGHT JOIN	INNER JOIN
34	3	Can you sort a joined result set by multiple columns at the same time using ORDER BY?	Yes, by separating the column names with commas.	No, ORDER BY only accepts a single column name.
35	3	What is the main structural purpose of using table aliases (like 'FROM employees e') when writing queries with JOIN statements?	To make the query run significantly faster in the database engine.	To shorten the query and distinguish between columns with identical names across different tables.
46	102	Which keyword is used to declare an integer variable?	int	num
47	102	Which data type is used to store decimal numbers with higher precision?	char	double
48	102	Which operator returns the remainder of a division?	/	%
49	102	Which data type stores a single true/false value?	bool	char
50	102	What is the result of 7 / 2 in C++ when both operands are int?	3.5	3
51	102	Which operator is used to compare two values for equality?	=	==
52	102	Which data type is used to store a single character?	char	string
53	102	What does the assignment operator (=) do?	Compares two values	Assigns a value to a variable
54	102	Which of the following is a valid C++ variable name?	2ndValue	_total
55	102	What is the increment operator in C++?	++	--
236	4	What is a database?	A hardware component	An organized collection of data
237	4	Which component manages the database?	Operating System	DBMS
238	4	What is data redundancy?	Storage efficiency	Data duplication
239	4	Which model uses tables?	Hierarchical	Network
240	4	What is the function of a Primary Key?	To store images	To uniquely identify a row
241	4	What is a Database Schema?	The actual data	The design/structure
242	4	Which is a DBMS software example?	Microsoft Excel	MySQL
243	4	What is data integrity?	Data accuracy and consistency	Data volume
244	4	Who is the Database Administrator (DBA)?	User	Designer

Question_ID	Quiz_ID	Question_Text	OptA	OptB
245	4	What does DDL stand for?	Data Definition Language	Data Direct Logic
256	5	Which normalization form requires the removal of partial functional dependencies?	1NF	2NF
257	5	A table is in 3NF if it is in 2NF and has no ____ dependency.	Transitive	Partial
258	5	What is a functional dependency?	Relationship between attributes	A type of join
259	5	Normalization is primarily used to ____ anomalies.	Create	Reduce
260	5	Which anomaly occurs when we cannot insert data without other data?	Insertion anomaly	Deletion anomaly
261	5	In 1NF, an attribute must contain only ____ values.	Atomic	Composite
262	5	Which key is used to define the relationship in 2NF?	Primary Key	Candidate Key
263	5	Normalization helps to improve database ____ efficiently.	Integrity	Redundancy
264	5	What is the main characteristic of 3NF?	Removal of transitive dependency	Removal of primary key
265	5	Which of the following is a type of database anomaly?	Update anomaly	Search anomaly
266	101	Who created the C++ programming language?	Dennis Ritchie	Bjarne Stroustrup
267	101	Which header file must be included to use standard input and output streams like cin and cout?	<iostream>	<stdio.h>
268	101	Which operator is used to insert data into an output stream (like cout)?	>>	<<
269	101	Every statement in a C++ program must end with which character?	A period (.)	A colon (:)
270	101	Which data type is used to store a single character, such as 'A' or 'z'?	int	string
271	101	What is the correct way to start a single-line comment in C++?	//	/*
272	101	Which function serves as the entry point where every C++ program begins execution?	start()	main()
273	101	What is the result of the arithmetic expression 11 % 3 in C++?	3	2
274	101	Which keyword is used to declare a variable whose value cannot be changed later in the program?	static	final
275	101	What type of programming approach does C++ support?	Procedural only	Object-Oriented only
276	103	What is the standard library object used for reading user input from the console in C++?	cout	cin
277	103	Which loop is guaranteed to execute its body at least once, even if the condition is false from the start?	for loop	while loop
278	103	What keyword is used to immediately terminate the current loop or switch statement?	continue	break
279	103	Consider the loop: for(int i = 0; i < 5; i++); What is the output of this loop?	0 1 2 3 4	0 1 2 3 4 5
280	103	Which statement skips the remaining code in the current iteration and forces the next iteration of the loop to begin?	break	continue
281	103	Which looping construct is most appropriate to use when you know the exact number of iterations beforehand?	for loop	while loop
282	103	In an interactive menu-driven program (where a menu continually repeats until the user selects "quit"), which loop is the most natural and standard choice?	for loop	do-while loop
283	103	What will happen if a loop's condition evaluates to true and there is no way for the condition to ever become false?	A compilation error	An infinite loop
284	103	Which of the following is the correct extraction operator used with cin to get user input?	<<	>>
285	103	What happens if a user enters a string (like "hello") when a cin >> variable; statement is expecting an integer?	The variable is automatically converted into a string.	cin enters a fail state, skips future inputs, and the loop can become infinite.
286	201	Which component of the CPU is responsible for performing arithmetic and logical operations?	Control Unit	Arithmetic Logic Unit (ALU)
287	201	What does the acronym "RAM" stand for in computer hardware?	Readily Access Module	Random Access Memory
288	201	Which of the following is considered the absolute fastest memory in a computer's hierarchy?	Cache memory	Main memory
289	201	What is the fundamental electronic circuit used to store a single bit of data?	Multiplexer	Flip-Flop
290	201	Which type of bus is used to carry the physical address of the memory location that the CPU wants to access?	Data bus	Address bus
291	201	The standard, widely-used modern computer structure where data and instructions are stored in the same memory unit is known as:	Harvard Architecture	Von Neumann Architecture
292	201	Which architecture features a large and complex set of instructions, aiming to execute high-level operations directly in a few lines of code?	RISC	CISC
293	201	What does the acronym "CPU" represent?	Central Processing Unit	Computer Processing Unit
294	201	What is a Memory Address Register (MAR) used for?	Storing the data currently being read from memory	Storing the address of the memory location being accessed
295	201	Which of the following is an example of non-volatile memory?	RAM	Cache memory
296	202	What does the acronym ISA stand for in processor architecture?	Instruction System Algorithm	Instruction Set Architecture
297	202	Which of the following is a primary characteristic of a RISC (Reduced Instruction Set Computer) processor?	It has a large, complex set of instructions.	It executes instructions using a simple, power-efficient set.
298	202	Which processor architecture approach uses very long instruction words to perform multiple operations in parallel?	CISC	VLIW
299	202	What does the term "pipelining" primarily aim to improve in a CPU?	Throughput by allowing multiple instructions to overlap in execution	Memory capacity
300	202	Which of the following is considered a primary classification of a processor's microarchitecture?	CISC	Compiler
301	202	What is a "data hazard" in processor pipelining?	When the processor runs out of electrical power.	When an instruction depends on the result of a previous instruction that has not yet been completed.
302	202	Which architecture features separated memory pathways for instructions and data?	Harvard Architecture	Von Neumann Architecture
303	202	What is the standard measurement for CPU execution speed, calculated as the number of instructions executed per second?	MIPS	RPM
304	202	In modern microprocessors, what does a "branch prediction" unit do?	It calculates memory addresses.	It guesses the outcome of a conditional jump to prevent pipeline stalls.
305	202	Which processor architecture technology splits a single physical CPU core into two virtual cores to improve task multitasking?	Hyper-Threading	Multi-Core

Question_ID	Quiz_ID	Question_Text	OptA	OptB
306	203	Which memory type is located closest to the CPU and operates at the highest speed?	Main Memory (RAM)	Cache Memory
307	203	What is the primary purpose of Cache Memory?	To increase the permanent storage capacity of the computer	To bridge the speed gap between the CPU and main memory
308	203	In cache memory mapping, which technique maps any block of main memory to any line in the cache?	Direct Mapping	Set-Associative Mapping
309	203	What technique divides memory into fixed-size blocks to translate virtual addresses into physical addresses?	Segmentation	Paging
310	203	Which mapping technique restricts a main memory block to map onto a specific, fixed line in the cache?	Direct Mapping	Fully Associative Mapping
311	203	What does the acronym "TLB" stand for in virtual memory systems?	Transfer Lock Buffer	Translation Lookaside Buffer
312	203	Which memory is volatile and loses its data when the power is turned off?	ROM (Read-Only Memory)	RAM (Random Access Memory)
313	203	What happens during a "Cache Miss"?	The requested data is found in the cache	The requested data is not found in the cache and must be fetched from main memory
314	203	Which memory has the largest capacity but the slowest access time in a standard computer hierarchy?	CPU Registers	Cache Memory
315	203	What does the "Principle of Locality" in memory systems state?	Programs tend to access a localized, predictable portion of their memory address space at any given time	Memory chips must be placed physically close to the CPU on the motherboard

Column	Type	Null	Default	Comments
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Quiz_ID	int(10)	No		
attempt_no	int(11)	No	1	
Grade	int(3)	No		
date_taken	datetime	Yes	current_timestamp()	

userID	Quiz_ID	attempt_no	Grade	date_taken
D032410025	1	1	90	2026-07-04 21:45:37
D032410025	2	1	90	2026-07-04 21:45:37
D032410025	3	1	90	2026-07-04 21:45:37

Column	Type	Null	Default	Comments
Subject_Code	varchar(20)	No		
Title	varchar(100)	No		
Description	varchar(255)	No		

Subject_Code	Title	Description
DITP1113	Programming 1	Code Language C++ Parameter
DITP2913	Database	Databases are structured to facilitate storage, retrieval, modification, and deletion of data in conjunction with various data-processing operations.
DITS1133	COA	Data for COA

Column	Type	Null	Default	Comments
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Email	varchar(100)	No		
Password	varchar(255)	No		
Phone	varchar(20)	No		
Role	varchar(20)	No		
user_status	int(1)	No	0	
Verification_Status	varchar(20)	No		
reset_token	varchar(255)	Yes	NULL	
token_expiry	datetime	Yes	NULL	
otp_code	varchar(6)	Yes	NULL	

userID	Name	Email	Password	Phone	Role	user_status	Verification_Status	reset_token	token_expiry	otp_code
A032410201	Ahmad Arfan Danish bin Dato' Arman	a032410201@admin.utem.edu.my	student123		Admin	0	ACTIVE			
D032410021	Siti Nurahaliza	ali@gmail.com	abc123		Student	0	ACTIVE			
D032410022	Muhammad Rizal bin Mohktar	rizal@gmail.com	Awa	0228932	Student	1				
D032410025	Noor Iman Nabil bin Roslan	d032410025@student.utem.edu.my	abc123		Student	0	PENDING			
D032410324	Ahmad Rosli bin Rizal	d032410324@student.utem.edu.my	student123		Student	0	ACTIVE			
D032410728	Nobruzane Smith Andre	d032410728@student.utem.edu.my	123456		Student	0	PENDING			
L1006001	Ts. Nor Mas Aina Md. Bohari	L1006001@lecturer.utem.edu.my	123		Lecturer	0	ACTIVE			
L1006789	Ts. Shah Idris bin Muhaimin	L1006789@lecturer.utem.edu.my	123		Lecturer	0	ACTIVE			
L1006893	Dr. Rina binti Zahrin	L1006893@lecturer.utem.edu.my	123		Lecturer	0	ACTIVE			

Column	Type	Null	Default	Comments
progressID	int(11)	No		
userID	varchar(12)	Yes	NULL	
Subject_Code	varchar(20)	No		
chapterIndex	int(11)	No		
completedAt	timestamp	No	current_timestamp()	

progress ID	userID	Subject_Code	chapterIndex	completedAt
1	D032410025	DITP2913	1	2026-07-02 00:05:41
2	D032410025	db	1	2026-07-02 11:30:29
3	D032410025	cpp	1	2026-07-02 11:59:31
6	D032410025	cpp	2	2026-07-02 02:55:38
20	D032410025	db	2	2026-07-02 12:41:01
30	D032410025	db	3	2026-07-02 12:34:06